

Life Cycle Assessment

Standardised accounting of environmental impact across a product's whole life

GLOSSARY 05 - SHEET 7 / 8

PROFESSIONAL DEFINITION

Life cycle assessment is the compilation and evaluation of the inputs, outputs and potential environmental impacts of a product system throughout its life cycle, from raw material extraction through production, use and end-of-life. It is standardised in four phases - goal and scope definition, inventory analysis, impact assessment, and interpretation - by ISO 14040 and ISO 14044, and its results are only comparable where the system boundary and functional unit are identically defined (ISO, 2006).

WHO DEFINED IT ORIGINALLY

The method emerged from packaging studies conducted for the Coca-Cola Company by the Midwest Research Institute in 1969 and was developed through the Society of Environmental Toxicology and Chemistry from the 1990s. It was standardised by the International Organization for Standardization as ISO 14040 and ISO 14044, first published in 1997 and revised in 2006.

WHY THIS DEFINITION WAS DEVELOPED

Environmental claims were being made on the basis of a single life-cycle stage, which permitted burden-shifting - a product could be made cleaner in use by being made dirtier in manufacture, and reported as an improvement. A standardised whole-life boundary made such shifting visible and made competing claims comparable.

FIGURE 1 - THE FOUR PHASES OF AN ASSESSMENT

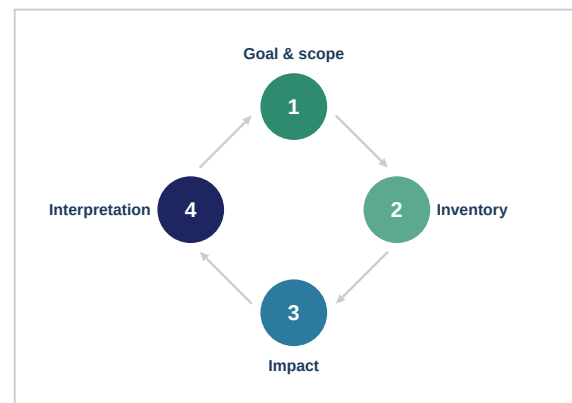


Figure 1. Interpretation feeds back into scope, which is why the method is iterative rather than linear. For AI systems the first phase is decisive: whether hardware manufacture and inference at scale sit inside the boundary determines the result more than the inventory does (after ISO, 2006).

HOW IT IS USED

Define the functional unit and system boundary before collecting any data, since these determine the result more than the inventory does; for AI systems this means deciding explicitly whether hardware manufacture, training and inference are inside the boundary.

WHEN IT IS USED

In sustainability reporting, in procurement of AI infrastructure, when comparing an AI-enabled process against the process it replaces, and when an environmental claim is made about a digital service.

BY WHOM IT IS USED

Sustainability and reporting functions, procurement and supply chain teams, environmental consultants conducting the assessment, and auditors verifying disclosed figures.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor practice was single-issue or single-stage assessment - energy in use, or emissions at the factory gate - which invited burden-shifting between stages and between impact categories. The ISO structure requires a declared functional unit and boundary and mandates the interpretation phase, in which sensitivity to those choices must be examined. For AI this is the decisive discipline, since assessments that exclude hardware manufacture or inference at scale can differ from complete ones by a large factor.

RELATED DEFINITIONS IN THIS COURSE

Green AI - Unit 5

Externality - Unit 5

AI Governance - Unit 6

Benefits Realisation - Unit 4

SOURCES (HARVARD REFERENCING STYLE)

International Organization for Standardization (2006) ISO 14040:2006 Environmental Management - Life Cycle Assessment - Principles and Framework. Geneva: ISO.

International Organization for Standardization (2006) ISO 14044:2006 Environmental Management - Life Cycle Assessment - Requirements and Guidelines. Geneva: ISO.

Hellweg, S. and Mila i Canals, L. (2014) 'Emerging approaches, challenges and opportunities in life cycle assessment', Science, 344(6188), pp. 1109-1113.